Domain IV Addendum: Stoichiometric & Hardware-Gated IT-OT Process Security (`OrgelKraft` / `SPEC-PODS-PENTAD-RB-12`)

Pre-Meeting One-Pager for SFI NORCICS & Norwegian Computing Center (Process Industry, Subsea & Grid Alignment)

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Context: Pre-meeting technical addendum extending `IT_OT_CONSEQUENCE_GATE_NORCICS_BRIEF.md` (Domains I–III) to **Domain IV: Electrochemical, Subsea & Process-Industry OT Security** (relevant to SFI NORCICS user partners such as Yara, Norsk Hydro, Equinor, Kongsberg, Elvia, and Sykehuset Innlandet)
Full Monograph (`v1.0.1`, 74 pp.): KONSEPTUELL_GRANITT_ET_FORSOK_PA_SANNHET.pdf
Epistemic Status: `L1` Deterministic Software, Exact Rational Ledger & RTL Simulation (`INSPIRED_BY != IMPLEMENTS != PROVES`; `L2/L3` physical reactors and subsea hardware held in explicit `HOLD` at 0.00 V DC pending HIL/field qualification)

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1. Architectural Motivation: Closing the Process-Control & Transducer Gap

In continuous chemical, electrochemical, and subsea process plants, Stuxnet-class IT-OT attacks and AI-driven optimization failures exploit three structural blind spots in classical SCADA/PLC and Simplex/SOTIF architectures:

1. **Floating-Point Stoichiometric Drift:** IEEE-754 floating-point tolerances allow covert mass-balance deviations, unrecorded feedstock injection, or gradual setpoint creep (e.g., pushing an electrolyzer above its water-splitting catalyst-degradation ceiling or driving a crystallizer across the Ostwald nucleation boundary) to hide inside numerical noise.
2. **The 400-Clock-Cycle Audit Window:** A `POSIX fsync()` disk commit requires ~ 5.0 ms, whereas an 80 kHz FPGA safety loop ticks every 12.5 μs. If a control system waits for disk logging before tripping—or logs intention before physical actuation—an adversary gains a **400-cycle window** to corrupt state before the relay opens.
3. **The Transducer & False-Realization Gap ($W \not\equiv$ software log):** Classical controllers log a command as "executed" once it leaves the digital output register ($Q_{\text{math}} \sim k_B T \ln 2$), remaining blind if a valve seizes mechanically downstream. Across 8,000 deterministic Monte Carlo fault-injection runs (`L1`, `seed=42`), classical command-based architectures exhibit a 25.00% **false-realization rate** (2,000/8,000), whereas the $T_{KY} = \Omega \circ \Pi_K \circ \Phi$ realization grammar achieves 0.00% **false positives** (0/8,000).

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2. Three `L1`-Verified Mechanisms in `OrgelKraft` (`SPEC-PODS-PENTAD-RB-12`)

A. Exact Rational Mass-Balance Compiler ($\Delta R = 0$ across 45 Conservation Checks)

Rather than approximating process flows in floating-point arithmetic, the `OrgelKraft` reference balance models a closed six-element cascade (C, H, N, O, Mg, P) using exact rational arithmetic (`fractions.Fraction`). Across a reference 24-hour cycle—spanning low-voltage nitrogen fixation (`K-POD`, parallel cell stack at $\leq$ 0.998 V DC, 2,204.41 A, 2,200 W), aromatic reduction (`C8` at 1.050 V DC), struvite phosphorus recovery (`P-POD` locked inside the Ostwald metastable window $S_{\text{struvite}} \in [1.80, 3.20]$), and enzymatic lignocellulose bioconversion (`POD-Mycelin` at 50.0 W±0.5 W)—total mass enters at 141.10099788325846 kg and exits at 141.10099788325846 kg with **zero rational residual ($\Delta R = 0$)**. Any unrecorded mass deviation exceeding 1 mg (`delta_q_mg > 1`) immediately trips the gate to 0.00 V DC.

B. 12.5 μs SystemVerilog Guillotine & Asymmetric Time-Domain Bifurcation

In the RTL reference modules `kpod_fpga_guillotine_12u5` and `etor_pik_boundary_gate`:

- **Memory & Fuse Integrity:** Every 12.5 μs clock cycle verifies an involutive XOR round-trip in reversible phase memory (`((payload_in ^ MASK) ^ MASK == payload_in`) and enforces a 64-bit monotone fuse register (`(F_n & F_{n+1}) == F_n`) that forbids state rollback.
- **Asymmetric 400-Cycle Rule:** For **admission (`OPEN` → 5.00 V DC)**, the sealed witness log is a mandatory entry ticket (the actuator waits all 400 cycles until `fsync` completes). For **protective cutoff (`KILL` → 0.00 V DC)**, causality reverses: the FPGA/silicon crowbar drops excitation to 0.00 V DC on **Cycle 1 (**12.5 μs **/** 4.0 ns**)** while the witness log is written in parallel as a post-mortem autopsy.

C. Two-Phase Verification (Π_K vs. Π_{orth}) & Anti-DoS Galvanic Separation

To prevent both sensor spoofing and biological alarm-fatigue Denial-of-Service (DoS):

1. **Phase A (Pre-Commit in Π_K) vs. Phase B (In-Flight in Π_{orth}):** Π_K verifies pre-commit viability (causal light-cone $\tau_{\text{ToF}} \leq 133.425$ ns cross-locked against the $c_{\text{acoustic}} \approx 11,100$ m/s sapphire crystal eigenfrequency within ±5 ppm, Nagumo tangency $\dot{x} \in T_K(x)$, and forward braking horizon across 41 discrete trajectories) *before* granting 5.00 V DC. Once energized, Π_{orth} monitors cross-modal physical realization (coil back-EMF $\int V(t)I(t)\,dt$, thermal enthalpy ΔT, and acoustic ToF) within a 38.4 ms actuation window, firing a secondary galvanic crowbar (`GATE :: KILL`) on any divergence.
2. **Electronic `CAUSAL_HOLD` (8.0 μJ) vs. Mechanical Trip-Free `RESET_BY_L01`:** External network/RF noise is rejected in the solid-state airlock (`CAUSAL_HOLD` at 0.00 V DC, 8.0 μJ per pulse, yielding a 1.25×10^9 thermodynamic exhaustion ratio against a 10 kJ attacker) without tripping mechanical latches. Irreversible spring-return `GATE :: KILL` is reserved strictly for internal core breaches and requires a two-stage mechanical **trip-free `Level 01` human reset** (where the first 45° of lever travel drives a shaft magneto to power a read-only Π_{orth} check alongside passive Bourdon/bimetallic anchors before copper contacts can latch).

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3. Summary Extension Matrix (Domain IV — SFI NORCICS Alignment)

SFI NORCICS Sector	Untrusted Candidate (Φ)	Deterministic Gate & Invariant (Π_K, Π_{orth})	Fail-Closed Verdict (Ω)	`L1` Verification Artifact
Electrochemical & Process OT *(Yara / Hydro)*	PLC/AI voltage, temperature or feedstock setpoint	Exact rational stoichiometry ($\Delta R = 0$, 45 checks), $V \leq 0.998$ V DC, Ostwald $S \in [1.80, 3.20]$, XOR & monotone fuse	12.5 μs RTL cutoff to 0.00 V DC on Cycle 1	`kpod_fpga` (12.5 μs) & `fractions` suite
Subsea & Kinematic Actuation *(Equinor / Kongsberg)*	Valve / actuator trajectory near structural limit Q_{LIMIT}	41-candidate braking horizon ($\lvert\mathcal{A}(x)\rvert \geq 1$) + 38.4 ms in-flight Π_{orth} (EM + ΔT + acoustic ToF)	Auto-reverse at ∂K; Crowbar `KILL` if actuator seizes	`etor_pik` gate (`seed=42`) & 8,000 MC runs
Grid, Police & Hospital Edge *(Elvia / Politiet / Sykehuset)*	Contested fiber/RF command or high-entropy 6G ISAC waveform	$\partial A / \partial C_{\text{fiber}} \equiv 0$, sapphire-locked $\tau_{\text{ToF}} \leq 133.425$ ns, Woodward $\chi(\tau, \nu)$ vs. $H(X)$	8.0 μJ `CAUSAL_HOLD` on channel noise; trip-free `L01` on core fault	Local-first SHA-256 ledger & ISAC covariance gate